LABSHEET-07

Q1. Display the processes associated with the current user using ps and observe the PID, terminal and process status.

```
ps
```

Q2. Display all processes running on the system using an appropriate ps option and identify the PID, user and command fields.

```
ps -ef
```

Q3. Display a detailed process listing using ps aux and identify the CPU and memory usage fields.

```
ps aux
```

Q4. Display the process hierarchy using ps with an option that shows parent-child relationships.

```
ps -ef --forest
# or
ps auxf
```

Q5. Start the top command and observe the running processes, CPU usage, memory usage and process IDs.

```
top
```

Q6. Use top to identify a process with high CPU or memory usage and record its PID.

```
top
# Press 'P' to sort by CPU usage or 'M' to sort by Memory usage, then note the PID.
```

Q7. Start htop and use its interactive interface to view processes and resource usage.

```
htop
```

Q8. Start a long-running command in the foreground and terminate it using Ctrl+C.

```
sleep 100
# Press Ctrl+C to terminate
```

Q9. Start a long-running command in the background using the & operator and display the assigned job information.

```
sleep 100 &
```

Q10. Display the jobs currently running in the shell using the jobs command.

```
jobs
```

Q11. Bring a background job to the foreground using the fg command and then terminate it safely.

```
fg %1
# Press Ctrl+C to terminate
```

Q12. Suspend a running foreground process using Ctrl+Z and verify that the job has been stopped.

```
sleep 100
# Press Ctrl+Z
jobs
```

Q13. Resume a stopped job in the background using bg and verify its status using jobs.

```
bg %1
jobs
```

Q14. Use jobs -l to display the job number and PID of background or stopped jobs.

```
jobs -l
```

Q15. Use kill with a selected PID to terminate a process and verify that the process has ended.

```
kill <PID>
ps -p <PID>
```

Q16. Use kill -TERM with a selected PID and explain why SIGTERM is normally preferred before forceful termination.

```
kill -TERM <PID>
# SIGTERM (Signal 15) allows the process to clean up resources, close files, and exit gracefully.
```

Q17. Use kill -KILL with a selected test process when normal termination does not work, and verify the result.

```
kill -9 <PID>
# or kill -KILL <PID>
ps -p <PID>
```

Q18. Use pkill to terminate a selected test process by its process name and verify that it has ended.

```
pkill sleep
pgrep sleep
```

Q19. Use ps, jobs and appropriate process-control commands to identify and manage multiple test processes safely.

```
sleep 200 &
sleep 300 &
jobs -l
pkill sleep
```

Q20. Perform a complete process-management workflow: inspect processes, start a foreground job, move a job to the background, suspend/resume it, terminate it with kill, and clean up all test processes.

```
sleep 500
# Press Ctrl+Z to suspend
bg %1
jobs -l
kill %1
pkill sleep
```